

Assignment 6

Assignment Title

Design and develop a Dynamic Web Application using Node.js, Express, and MongoDB
(Vehicle Marketplace – Buy & Sell Used Cars & Bikes)

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Objectives

- To develop a dynamic web application using Node.js and Express
 - To understand backend API development and routing
 - To implement user authentication using sessions
 - To integrate MongoDB for storing vehicle listings and users
 - To design a real-world vehicle marketplace system
 - To perform CRUD operations (Create, Read, Update, Delete)
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Outcomes

- Learned backend development using Node.js and Express
 - Understood REST API creation and routing
 - Implemented MongoDB database with Mongoose
 - Built a responsive UI for vehicle listings
 - Gained experience in authentication and session handling
 - Improved debugging and problem-solving skills
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Theory Concept in Brief

1. Frontend

The frontend is responsible for displaying data to users.

In this project, HTML, CSS, and JavaScript are used to create a simple and clean UI where users can:

- View vehicle listings
 - Add new listings
 - Navigate through the website
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2. Backend

The backend handles requests and business logic.

Using Node.js and Express:

- APIs are created for authentication and product listings
- Routes handle user login, logout, and listing operations
- Middleware is used for authentication

As seen in your backend code:

- `/api/auth` → handles login/register
 - `/api/products` → handles vehicle listings
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3. Database

MongoDB is used as a NoSQL database to store:

- User details
- Vehicle listings

Mongoose is used to define schemas and interact with the database.

Example fields stored:

- Title, brand, model, price
 - Year, fuel type, kilometers driven
 - User who posted the listing
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Technologies Used

Frontend Technologies

- HTML → Structure
 - CSS → Styling
 - JavaScript → Functionality
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Backend Technologies

- Node.js → Runtime environment
 - Express.js → Web framework
 - Express Session → Authentication
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Database

- MongoDB → NoSQL database
 - Mongoose → ODM
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Core Functionalities Implemented

Vehicle Marketplace System

- User Registration and Login
 - Session-based Authentication
 - Add Vehicle Listing (Car/Bike/Scooter)
 - View Available Listings
 - Display Seller Information
 - Store Data in MongoDB
 - Dynamic UI Rendering
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Features from Your Project UI

(From your screenshot)

- Homepage with heading:
“Buy and Sell Used Cars & Bikes”
- Vehicle cards displaying:

- Name
- Year
- Price
- Seller

Examples:

- Renault Kwid (2017)
- Kia Seltos (2021)
- Yamaha FZ-S (2020)

Methodology / Steps Followed

1. Designed UI using HTML and CSS
2. Created Express server using Node.js
3. Connected MongoDB using Mongoose
4. Created User and Product models
5. Implemented authentication using sessions
6. Developed REST APIs for users and products
7. Created routes for adding and fetching listings
8. Seeded database with sample data
9. Displayed listings dynamically on frontend
10. Tested and debugged the application

Program Code (Overview)

Backend (Express Server)

- Server setup with middleware:
 - JSON parsing
 - Session handling
 - Static files
- Example:

```
app.use('/api/auth', authRoutes);
app.use('/api/products', productRoutes);
```

Database Seeding

- Sample users and vehicles inserted into MongoDB
- Links users with products

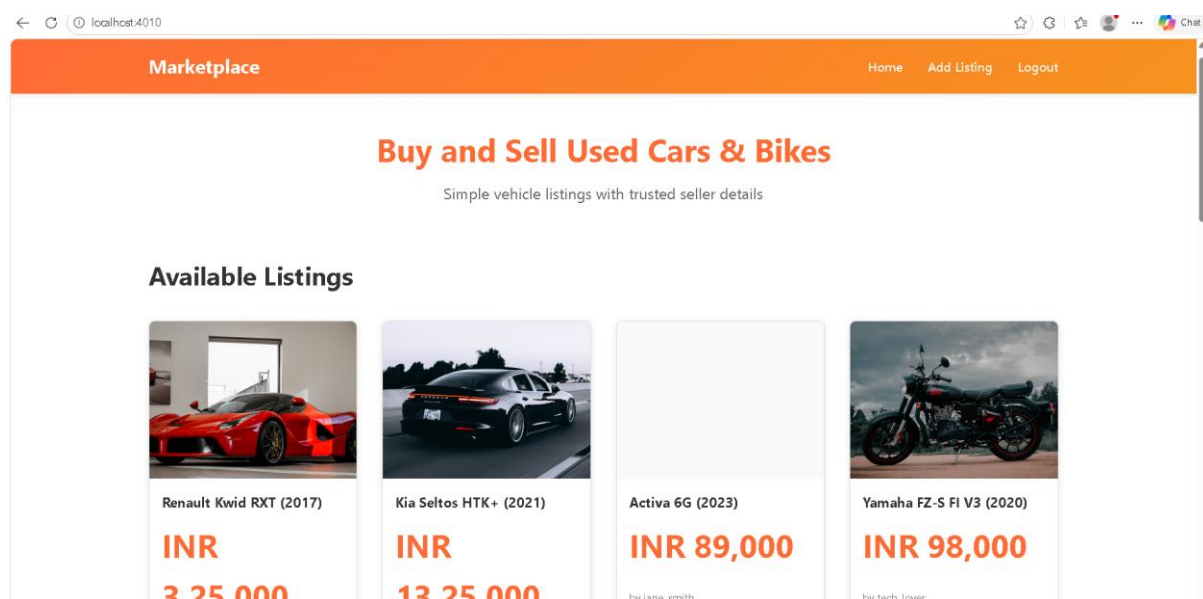
Example:

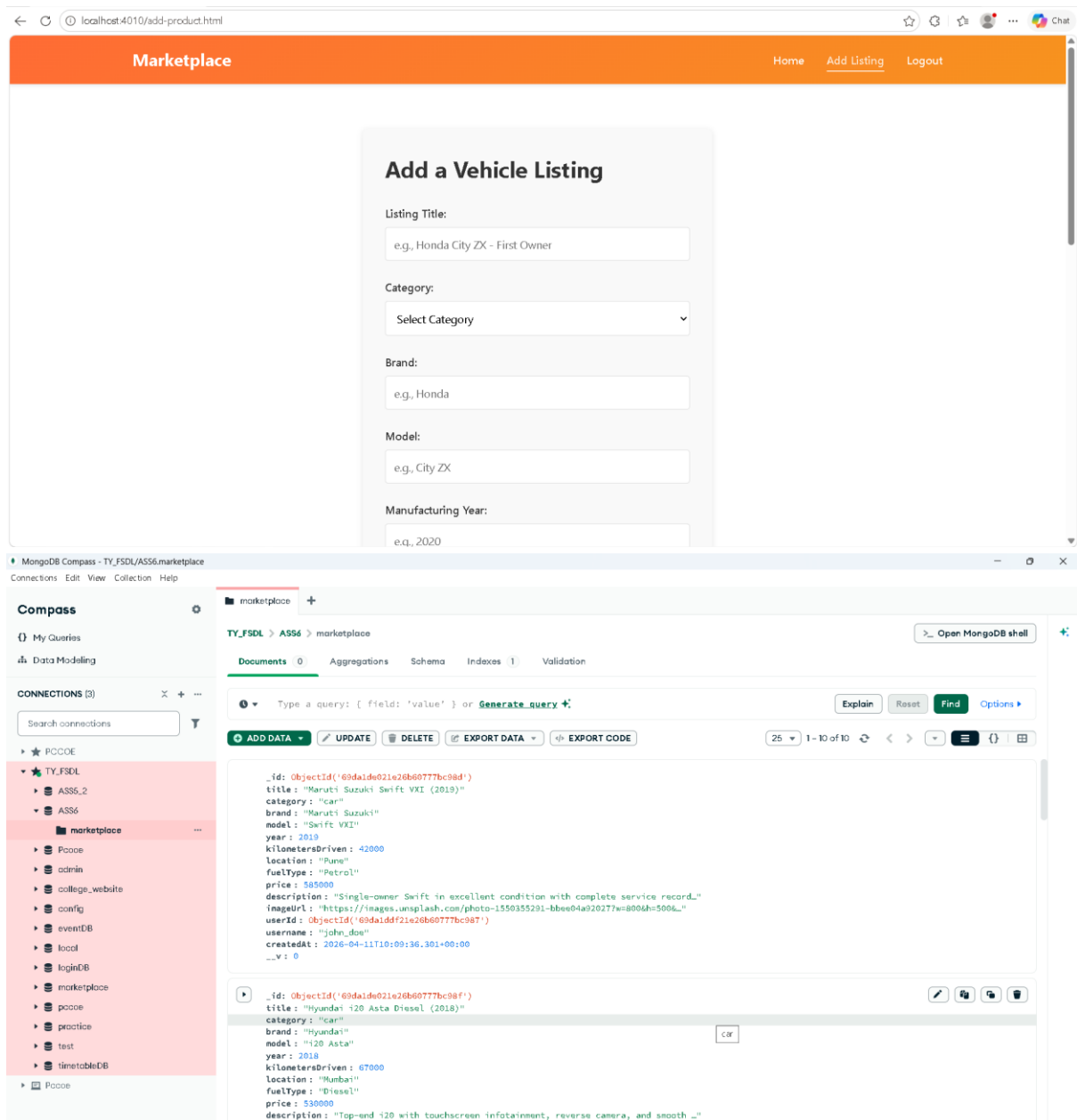
```
const product = new Product({
  ...productData,
  userId: user._id
});
```

Real-Life Applications

- OLX-like platforms
- Car resale websites
- Bike marketplaces
- Classified advertisement systems
- E-commerce platforms for used goods

Output Screenshots





Include:

- Homepage (Marketplace UI)
- Vehicle Listings Page
- Add Listing Page
- MongoDB Data (Compass Screenshot)

Your UI (as shown) includes:

- Clean marketplace layout
- Card-based listing system
- Pricing and seller details

Link to Codes

https://drive.google.com/drive/u/0/folders/1OkFcKX2y8e2Wjbb4_nxaoVWNzO-J7wKc

Conclusion

The assignment successfully demonstrates the development of a dynamic web application using Node.js, Express, and MongoDB.

The Vehicle Marketplace system allows users to buy and sell used vehicles with ease. It provides practical experience in backend development, API creation, database integration, and authentication.

This project enhances real-world development skills and prepares for building scalable web applications.